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Fast and Accurate Estimation of the HARDI Signal in Diffusion MRI Using a Nearest-Neighbor Interpolation Approach

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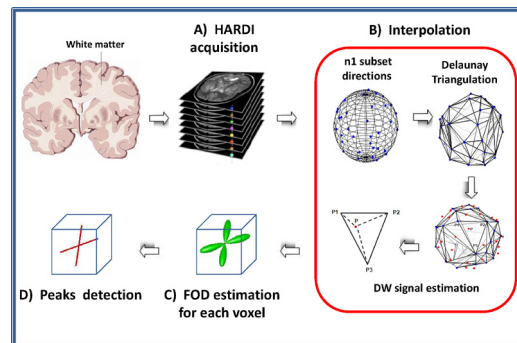
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Graphical abstract



Abstract

In the diffusion MRI domain, the HARDI methods were proposed to better characterize the complex biological tissues such as the white matter. In fact, they allow to overcome the problem of crossing fibers detection in the case of the Diffusion Tensor Imaging (DTI). However, the HARDI technique requires a very large number of Diffusion Weighted (DW) magnetic resonance images and thus it takes a long acquisition time, restricting its use in the clinical practice. We propose, in this paper, to develop a novel method for accelerating the reconstruction of the HARDI signal from a few number of DW images of the brain. The approach is a triangulation-based geometrical interpolation of existing signals. It consists on recovering non-acquired data according to their neighborhood from a reduced set of diffusion orientations on the sphere of the q-space. The accuracy of the proposed method was performed on two different phantom datasets for a qualitative and quantitative evaluation against known ground truths. We test also the robustness of the proposed approach under the noise. The obtained results demonstrate that the novel method can approximately halve the scan time and simultaneously obtain a proper fiber orientation estimation. In fact, the estimated FOD (e.g. FOD, Fiber Orientation Distribution) function based on only 34 directions is nearly identical to the one calculated with the full HARDI acquisitions (65 directions). Comparative evaluation on standard phantoms show that the proposed approach outperforms a large selection of methods of state-of-the-art according to the success rate and the angular error criteria.

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1. Introduction

Diffusion-Weighted MRI (DW-MRI) is the only non-invasive tool to reconstruct white matter fiber bundles in the brain and to quantify tissue changes in-vivo. In fact, the locations and the orientations of the fibers are determined indirectly by measuring the anisotropy of local water diffusion. The DW-MRI provides quantitative indices that are exploited by neurologists for the diagnosis of a wide variety of brain pathologies including infections, degenerative diseases and inflammatory conditions (multiple sclerosis) [1,2]. Another class of applications of the DW-MRI focus on the brain tractography. It is with a paramount importance for pre-operative planning before intracranial mass resection in the brain tumors [3,4]. It allows the anatomical study of fiber bundles that are in contact with the lesion.

Diffusion Tensor Imaging (DTI) [5] was the first proposed DW-MRI approach to characterize the anisotropic diffusion of water molecules inside the brain tissues. It captures one fiber direction per voxel and it fails in regions containing multiple fiber orientations due to its Gaussian diffusion assumption. The choice of local reconstruction methods is, therefore, crucial for discerning multiple fiber orientations.

In this context, a novel DW-MRI technique called Diffusion Spectrum Imaging (DSI) was proposed. This technique allows the construction of the Ensemble Average Propagator (EAP) with a discrete Fourier transform, from a Cartesian sampling of the diffusion signal [6,7]. The EAP provides the full information about the diffusion process. It is also possible to resolve intravoxel fiber crossing using the High Angular Resolution Diffusion Imaging (HARDI) [8]. It has become one of the most widely used methods owing to its ability to characterize the intra-voxel crossing fibers from the diffusion-weighted MRI data. It allows to obtain an accurate assessment about the neural architecture. This technique is based on sampling the signal on a single sphere of the q-space. Several HARDI-based methods have been proposed to address this issue. The Q-Ball Imaging (QBI) technique, proposed by Tuch et al. [9], reconstructs the Orientation Distribution Function (ODF) by applying the Funk Radon transformation of the diffusion signal. Descoteaux et al. [10] proposed the reconstruction of the diffusion signal by the spherical harmonics to obtain an analytical solution of the Radon transformation. Tournier et al. [11] proposed another kind of HARDI-based method. It consists on estimating the Fiber Orientation Distribution function (FOD) in each imaging voxel. The FOD function is the result of the deconvolution of the diffusion MRI signal and the single fiber response. It permits an accurate reconstruction of the local fiber populations.

Successful resolution of crossing fiber orientations within each voxel requires a very large number of DW images. It takes a long acquisition time that limits the applicability of HARDI technique in the clinical diagnosing. This is problematic for the

clinical studies involving children and people afflicted with certain diseases.

For this reason, q-space acceleration techniques were proposed to reconstruct the HARDI signal from limited samples using hemispherical encoding scheme [12–14]. Sparse coding techniques such as Compressive Sensing (CS) and sparse regularization were also proposed to reduce the acquisition time. The compressive sensing theory, proposed by Donoho [15], aims to reconstruct a signal from undersampled measurements acquired below the Shannon–Nyquist rate. The CS technique was applied in High Angular Resolution Diffusion Imaging (HARDI) to reduce the acquisition time. Michailovich et al. [16] used this technique to construct highly sparse representation of the HARDI signals using the basis of spherical ridgelets.

Mani et al. [17] used also compressed sensing to recover the ODF function with combined k-space and q-space and with Total Variation regularization in the spatial domain. Some approaches have been recently proposed in order to build dictionaries that enable sparse representations of the DW-MRI signal [18]. Tristán-Vega et al. [19] focused on the orientation distribution function (ODF) estimation in the frame of Spherical Wavelets (SW) using sparse regularization. Daducci et al. [20] used the sparse regularization and convex optimization in order to more optimally characterize the FOD sparsity. Auría et al. [21] proposed a novel algorithm to compute the FOD function based on spatial regularization directly on the fiber space.

In the CS technique, the random sampling scheme is used as a standard protocol for signal recovery. This fact can lead to a lack of information and does not ensure a good reconstruction in every case [22].

Accordingly, the present paper carries out the proposal and the validation of a novel method that reduces the acquisition time of the HARDI technique. It is based on a geometrical approach to construct the diffusion signal from the reduced number of gradient directions. These missing gradient directions will be obtained according to their neighborhood from a reduced set of data on the sphere of the q-space. We adopted standard metrics widely used in the literature, to assess the fiber orientation distribution (FOD) function reconstruction. The experimentations will be performed on two synthetic noisy diffusion MRI phantoms for a qualitative and quantitative evaluation of the proposed approach.

Thus, this paper will be structured as follows: we present in the second section a brief recall of the principle of the HARDI technique and the FOD function estimation. We detail all the steps of the proposed approach implementation in the third section. In the fourth section, we test its accuracy to perform a good estimation of the FOD by comparing the obtained reconstruction against the ground-truths via two main criteria. We also evaluate the effectiveness of our proposed approach in comparison with a large selection of methods from the state-of-the-art.

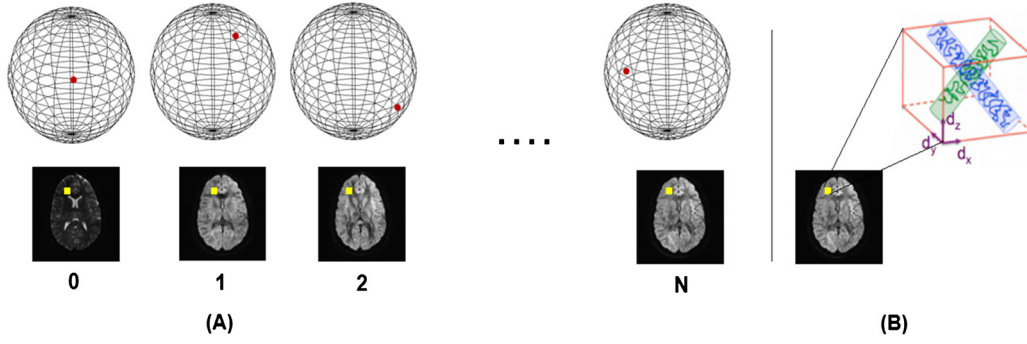


Fig. 1. The principle of the HARDI technique: (A) the coordinates of the wave vector in the q-space (top) and the corresponding diffusion weighted image (Bottom), (B) the 3D diffusion of water molecules within the voxel.

2. Background

2.1. The HARDI technique description

The diffusion is a random movement which is the result of thermal agitation. In the fluid such as CSF (Cerebral Spinal Fluid), the movement probability is the same in all directions. The movement is called, therefore, an isotropic diffusion. However, the diffusion may have a preferred direction and the probability is greater along a given axis resulting to the diffusion anisotropy.

Diffusion MRI is based on the application of a magnetic field gradient that sensitizes the effect of diffusion on the signal MRI amplitude. A diffusion gradient is symbolized by the q parameter representing the wave vector of displacements. It depends on the length, the separation, the strength, and the direction of gradient pulses in the imaging sequence [23]. By varying the applied diffusion-sensitizing gradients, we obtain series of Diffusion Weighted (DW) images. Every sampling point in the q-space (the space of all possible 3D q-vectors) corresponds to a specific direction and produces one Diffusion Weighted (DW) image.

High angular resolution diffusion imaging (HARDI) has become a very successful reconstruction technique that measures the water diffusion in a complex biological tissue such as the white matter along the N uniformly distributed directions on a single sphere of the q-space. Fig. 1 shows a graphical example of the principle of the HARDI acquisition procedure.

We first quickly review the principle of the used algorithm to generate the gradient directions. In HARDI technique, the sampling directions should be properly arranged as uniformly as possible on the sphere with antipodal symmetry, to recover accurate diffusion characteristics. This is done through an Electrostatic Energy Minimization (EEM) method, which was introduced in DW-MRI by Jones et al. [24]. It is the most popular way to uniformly distribute N points on a sphere. EEM algorithm considers the points as electrostatic charges, repulsing each other, and estimates the sample configuration by minimizing the electrostatic repulsion energy between two points q_i and q_j .

$$\min_{\{q_i\}_{i=1}^K} \sum_{j < i} \frac{1}{\|q_i - q_j\|_2^2} + \frac{1}{\|q_i + q_j\|_2^2} \quad (1)$$

The energy function can be approximately minimized incrementally [25]. K iterations are required and at each iteration i , we find the direction q_i minimizing the energy, while the $i - 1$ directions remain fixed. The minimization at each iteration is done by an exhaustive search among a set of 10000 pseudo-random points uniformly distributed on the sphere. This incremental construction of the uniform point set was useful in clinical scenario when the patient requests to abort the acquisition. The acquired orientation set will be fairly uniform and the acquisition can be partially saved.

The diffusion-weighted signals, in each voxel, are obtained by collecting the signal from the corresponding voxels in all the diffusion weighted images. It is defined as follows:

$$\text{Signal} = \{E_1, E_2, E_3, \dots, E_N\} \quad (2)$$

such as

$$E_i = S_i / S_0 \quad (3)$$

where E_i is the normalized signal, S_i is the measured signal obtained from the DW_{image_i} and S_0 is the unweighted one obtained without any applied diffusion gradients.

After recovering the HARDI signal, the next step consists on the estimation of the spherical function that characterizes the intra-voxel fiber orientation.

In this work, we focus on the spherical deconvolution methods currently used in DW-MRI to extract the FOD from the DW signal in each voxel. We present, in the following, a brief recall of the FOD function estimation.

2.2. The FOD function estimation

Spherical deconvolution techniques use the HARDI acquisition protocol to extract more information from the diffusion-weighted (DW) signal. The Fiber Orientation Distribution (FOD) function is estimated within each voxel directly from the diffusion-weighted (DW) data, using the concept of Spherical Deconvolution (SD) [11]. In this method, the HARDI signal is supposed to be the convolution of a single fiber signal and a function that characterizes the fiber orientation. Therefore, the FOD function is the result of the deconvolution of the diffusion signal and the single fiber signal. Fig. 2 illustrates the principle of this method.

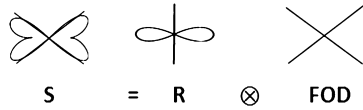


Fig. 2. The convolution of the fibers distribution with a single fiber R to compute the diffusion signal S as in [11] (Fig. 1).

The SD approach is linear and uses the spherical harmonics and the rotational harmonics to parameterize signal and the fiber response function, respectively.

Like most deconvolution problems, the spherical deconvolution problem is susceptible to noise and thus resulting spurious lobes. Thus, the original SD method of Tournier et al. [11] is improved by using the regularization which is, then, referred as a Constrained Spherical Deconvolution (CSD) method, to remove negative peaks in the FOD [26].

The used deconvolution method (CSD) iteratively imposes a non-negativity constraint for a set of control points. First, an initial estimation of the FOD is obtained. A set of 300 uniformly distributed directions is then identified, along which the FOD amplitude is negative (these directions were generated using the same electrostatic repulsion).

Using Tikhonov regularization [27] the FOD amplitude along these orientations is constrained to zero. An improved estimate of the FOD is then obtained by solving the Tikhonov problem, providing a new set of negative amplitude directions. This procedure is performed iteratively until convergence is achieved. A maximum harmonic degree of 8 ($L_{\max} = 8$) is generally recommended for the spherical harmonics (SH) representations when performing the CSD [28].

3. Proposed approach

We propose, here, an accelerated acquisition process in the diffusion MRI. Let us denote by N the full number of the used directions to construct the HARDI signal. These N directions are materialized by N points on the unit sphere of the q -space as explained in section 2 (Background). We propose to construct this HARDI signal by only a subpart of the full number of directions. We, therefore, use a subset of the N directions that we denote by n_1 . We intend to estimate the diffusion weighted signal of the n_2 ($n_2 = N - n_1$) samples to complete the full N directions. Fig. 3 illustrates the partial original signal (n_1 set) and the signal to be recovered (n_2 set).

The gradient table of the N directions uniformly distributed on the sphere of the q -space [29] is generated by the web application¹ based on the EEM algorithm described in section 2. It should be noted that this application generates incremental acquisition schemes. Thus, the subset (n_1) of the N directions is also uniformly distributed on the sphere. We propose a nearest neighbor linear interpolation approach of the DW signal values of the n_1 points on the unit sphere to estimate the DW signal of the n_2 set. For each point of the n_2 set, its neighboring points from the n_1 set are determined. Indeed, in the n_1 set, the DW signal values are known. The DW signal value of each point

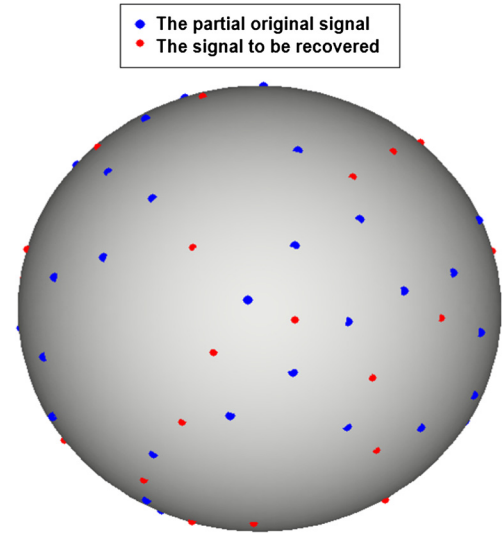


Fig. 3. Representation of the partial original signal (n_1) and the data to be recovered (n_2) on a sphere of the q -space.

of the n_2 set is estimated as a linear combination of the DW signal values of its neighborhood. We provide in the following the main steps of the proposed approach that models and estimates the diffusion signal from a limited number of samples (n_1 set). The proposed method can be summarized on two steps.

3.1. Step 1: Determination of the neighborhood of each point of the n_2 set

A subset of n_1 diffusion gradient directions with a nearly uniform coverage on the sphere was selected from the full set of the N gradients directions. We recall, here, that a DW signal is a continuous one that was sampled on the unit sphere [30]. Therefore, each point depends on its neighboring. We intend to estimate the DW signal values of each point of ($n_2 = N - n_1$) according to its neighboring points from the n_1 points set. For an accurate estimation, the neighborhood of a point P from n_2 should have a very near geometrical locations around the same point. Moreover, each point from the n_2 set should have the same neighboring points number. We propose, in this work, to use the Delaunay triangulation of the n_1 set of points (as illustrated in Fig. 4) in order to determinate the neighboring points of the n_2 points set. Indeed, each point P will have a nearest triangle from the Delaunay triangulation. The points composing this triangle correspond to the neighboring points of P .

The choice of the Delaunay triangulation can be justified with two main reasons:

- For each triangle from the Delaunay triangulation, there is no points inside its circumcircle. Therefore, when a triangle is chosen as a neighborhood of a point P , it covers locally an optimal neighborhood of this point.
- The Delaunay triangulation maximizes the minimum angle of all the triangle angles in the triangulation. We, therefore,

¹ <http://www.emmanuelcaruyer.com/q-space-sampling.php>.

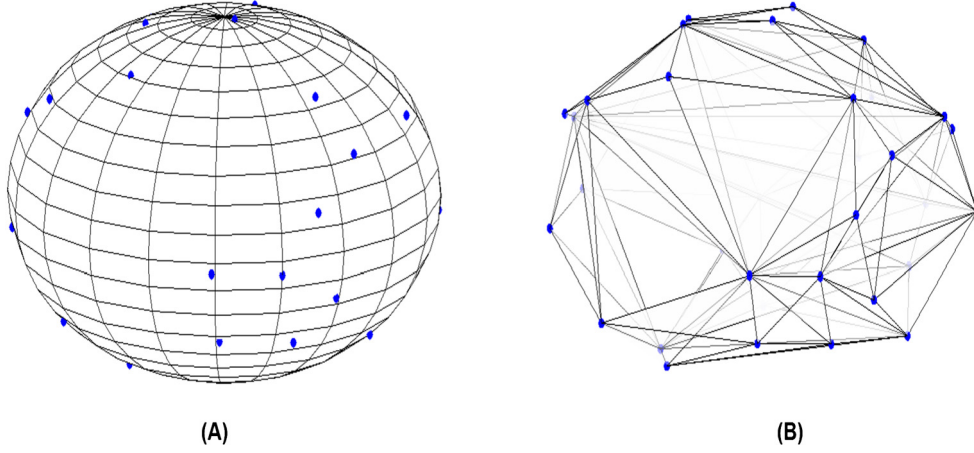


Fig. 4. The Delaunay triangulation: (A) the n_1 diffusion gradient directions, (B) the Delaunay triangulation of the original signal (the n_1 set of points).

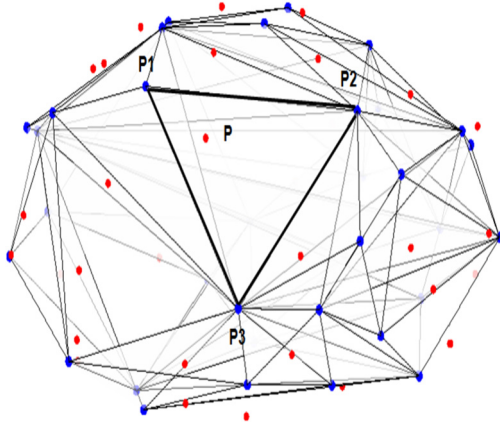


Fig. 5. The nearest triangle to a point P from (n_2) .

avoid triangles with one or two extremely acute angles that have undesirable properties in the interpolation process.

For a point P of the n_2 set, its nearest triangle from the Delaunay triangulation will be selected. The points composing this nearest triangle will be used as neighboring points of the points P . The problem of the neighboring points is transformed, therefore, to the one of the nearest triangle from the set of triangles obtained by the Delaunay triangulation. This nearest triangle corresponds to the one that has the minimum Euclidean distance between the considered point of n_2 and all the triangles obtained after the Delaunay triangulation.

Fig. 5 shows the nearest triangle to a point P from (n_2) . We obtain finally for each point of the n_2 set, its nearest triangle from the Delaunay triangulation.

3.2. Step 2: Estimation of the DW signal

Let P be a point of the n_2 set and P_1 , P_2 and P_3 three points from the n_1 set that compose the nearest triangle to P as illustrated in Fig. 6. We propose to estimate the DW signal value of the point P that we denote by $DW(P)$ as a linear combination of the DW signal values of the points P_1 , P_2 and P_3 .

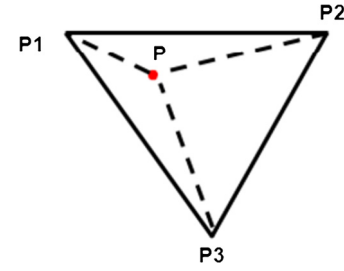


Fig. 6. An illustration of a point P (of n_2) and its nearest triangle from the Delaunay triangulation.

$$DW(P) = \alpha_1 DW(P_1) + \alpha_2 DW(P_2) + \alpha_3 DW(P_3) \quad (4)$$

α_1 , α_2 and α_3 are three real values that belong to $[0, 1]$ such that $\alpha_1 + \alpha_2 + \alpha_3 = 1$. They characterize the participation of each point from $\{P_1, P_2, P_3\}$ in the estimation of the DW signal in the point P . The closer the point P to a point $P_i \in \{P_1, P_2, P_3\}$ is, the larger the value of α_i must be. We are inspired for the computation of the values of $\{\alpha_i\}_{i \in 1..3}$ from the **barycentric coordinates of a point** in a triangle. Let $P_1 P_2 P_3$ denotes a triangle and P a point of this triangle. The coordinates of the point P in the barycentric coordinates system (P_1, P_2, P_3) are $(\alpha_1, \alpha_2, \alpha_3)$ such that:

$$\alpha_1 \overrightarrow{PP_1} + \alpha_2 \overrightarrow{PP_2} + \alpha_3 \overrightarrow{PP_3} = \vec{0} \quad (5)$$

α_1 , α_2 and α_3 are computed as follows:

$$\alpha_i = \frac{\text{Area}(P P_j P_k)}{\text{Area}(P_i P_j P_k)} \quad (6)$$

$$\alpha_1 + \alpha_2 + \alpha_3 = 1.$$

We can note here that the closer the point P to a vertex P_i of the triangle $P_1 P_2 P_3$ is, the larger the value of α_i is.

Following the same way as in the barycentric coordinates determination in a triangle, we estimate the values of $\{\alpha_i\}_{i \in 1..3}$ such as $DW(P) = \alpha_1 DW(P_1) + \alpha_2 DW(P_2) + \alpha_3 DW(P_3)$. In our context, all the points are a part of the sphere of the q space. Therefore, a point P does not belong obligatory to the plane of its nearest triangle $P_1 P_2 P_3$. Thus, in order to obtain a normalized values of $\{\alpha_i\}_{i \in 1..3}$, they are formulated as follows:

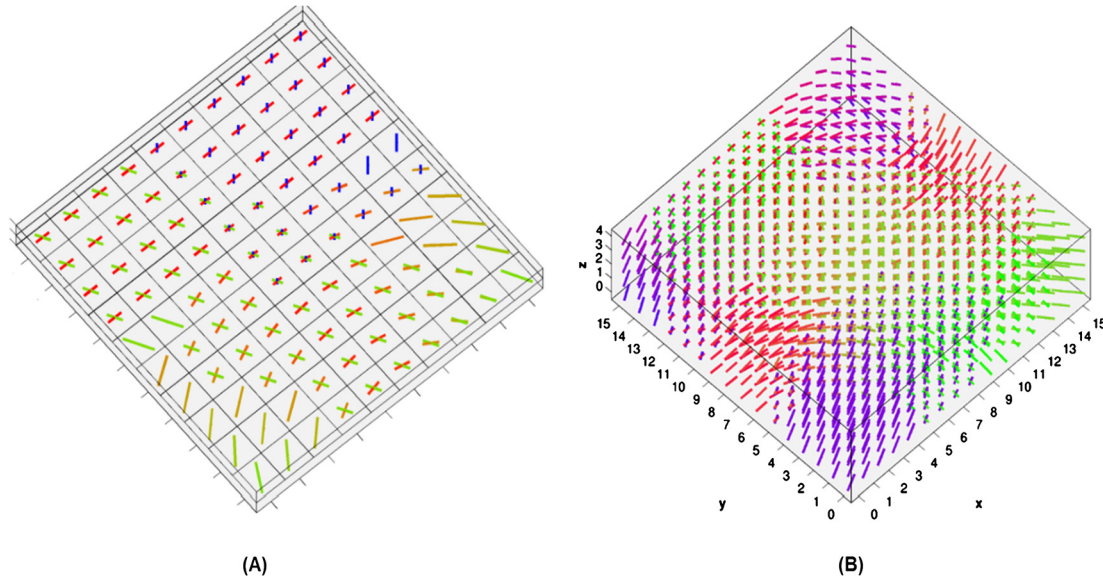


Fig. 7. The synthetic phantoms: (A) the 2D Structured Field synthetic phantom, (B) the 3D volume representing the Structured Field synthetic phantom.

$$\alpha_i = \frac{\text{Area}(P P_j P_k)}{\text{Area}(P P_i P_j) + \text{Area}(P P_i P_k) + \text{Area}(P P_j P_k)} \quad (7)$$

After recovering the missing data with the proposed method, we obtain a new diffusion signal composed by the n_1 directions and the recovered data (n_2 directions).

A partial part of this work was recently presented in [31]. We varied the subsets n_1 from 26 to 40 in order to evaluate the impact of the number of samples. After recovering the missing data with the proposed method, we obtain a new diffusion signal composed by the n_1 directions and the recovered data. Then, we calculated the fiber orientation distribution (FOD) using CSD ($L_{\max} = 8$). We reconstruct the FOD function from HARDI data with the novel estimated signal. The obtained FOD with the full original set directions (the N original values of DW signal) is used as a ground truth. According to the results of this study, it was proved that with 34 directions from 65, we obtain good results in terms of the angular resolution and the number of distinct fiber orientations.

In this work, we propose to perform an intensive experimentation and evaluation procedure in order to validate the proposed approach. We use in this context the well known gold standard database for this kind of experimentation.² The novelties of this work can be, therefore, summarized as follows:

1. The number of the crossing fibers is not limited to two. The context of the experimentation is much more difficult. This number varies from 1 to 4. We test, therefore, the power of the proposed approach for the determination of the crossing fibers number.
2. The angular resolution is not limited to 90° , 70° and 50° contrary to the previous work. Here, we will test the accuracy of the proposed approach for the good detection of angles between crossing fibers.

3. We test the robustness of the proposed approach against noise. Thus, in our work the diffusion-weighted data were generated for three signal-to-noise ratios ($\text{SNR} = 10, 20$, and 30).
4. We also compare the effectiveness of our approach with a large selection of methods participating in the HARDI reconstruction challenge 2012 using the same phantom data and the same evaluation criteria (Success Rate and the Angular Error).

4. Experimentation

We focus, in this section, on the experimentation procedure. We present in a first part the database we used and the evaluation criteria to test the performance of the proposed approach via the estimation of the FOD function. We expose in a second part, the obtained results according to the evaluation criteria. Finally, we compare the proposed approach with a large selection of methods of state-of-the-art.

4.1. Database and evaluation criteria

4.1.1. Description of the database

Since the ground truth of the in-vivo human data is not available, the human data can be only analyzed visually. Numerical simulations offer a great way to assess the performances of the proposed approach for a range of experimental conditions: the number of distinct fiber orientations, the crossing angle between the fiber compartments and the noise levels ($\text{SNR} = 10, 20$, and 30). In this study, we test the proposed approach on two kinds of synthetic diffusion MRI phantoms with known ground-truths. These phantoms were developed for the HARDI reconstruction Challenge 2012, Workshop. The first synthetic phantom consists in a 2D slice as shown in Fig. 7(A). It is stored as a 10×10 cells array containing each one a different number of fiber populations (from one to three).

² <http://hardi.epfl.ch/static/events/2012ISBI/>.

The second synthetic phantom is a 3D volume with $16 \times 16 \times 5$ voxels that comprises five different fiber bundles resulting in voxels with bending, crossing and kissing tracts. Tournier et al. [32] prove that more than the minimum of 45 DW directions must be used to ensure a good uniformity of the DW gradient directions with a maximum harmonic degree of 8 ($L_{\max} = 8$) and for a b -value equal to 3000 s/mm^2 . Most of works in the literature used a number of directions greater than 60 DW directions [33,34]. Thus, in our work Diffusion-weighted data were generated assuming a typical 65 direction sampling scheme and for three signal-to-noise ratios (SNR = 10, 20, and 30). For each voxel configuration, the diffusion weighted signal S along a given diffusion weighted gradient encoding axis was simulated following the multi-tensor model as in Tuch et al. [9]. Rician noise with different signal-to-noise ratios (SNR) (SNR = 10, 20, and 30) was, then, added to each DW signal. The first phantom permits, unlike the second one, a visual illustration of the ground truth.

Therefore, a visual comparison between the ground truth and the obtained results is possible. Simulations have been carried out to study the effects of reducing the number of samples (from 65 to 34) and to verify if and how the proposed approach is able to preserve results with these configurations. Finally, we analyze the influence of noise on the FOD estimation.

4.1.2. Evaluation criteria

In order to perform a quantitative evaluation of the computed FOD function from the recovered signal using the proposed approach, we use the following indices:

- The success rate: It corresponds to the percentage of voxels within which the reconstruction algorithm was able to correctly estimate the number of fibers populations.
- The angular error: It is assessed through the mean angular error (in degrees) between the extracted peaks and the true ones present in a voxel. Each estimated fiber orientation is paired with only one actual fiber orientation from the ground-truth. It is formulated as follows:

$$\theta = 180/\pi * \arccos(f_{\text{true}} \cdot f_{\text{estimated}}) \quad (8)$$

where f_{true} is a true fiber population, $f_{\text{estimated}}$ is the closest fiber to the estimated directions and $(\cdot)$ denotes the dot product.

4.2. Results and discussion

We test the performance of the proposed approach using two experiments: In the first one, we use the first 2D phantom to perform assessment by visual comparison with the ground-truth (qualitative evaluation) while the second experiment consists on a quantitative validation of our method using the second DW phantom.

4.2.1. Qualitative evaluation

In order to test the ability of our method to correctly detect the number of fiber orientations and the crossing angles,

we first perform a qualitative comparison using the synthetic data of the first phantom. We compute the FOD function, using the CSD method, based on the obtained HARDI signal using the proposed approach and we extract the local maxima of the FOD function in order to evaluate its accuracy. We, then, compare the estimated FOD with the corresponding ground truth and with the obtained FOD with the full original set directions (the N original values of the DW signal). Fig. 8 shows how the reconstruction obtained with the proposed approach is able to accurately estimate the unmeasured directions in the presence of noisy synthetic data (SNR = 30). Fig. 8(C) shows the ground truth. The FODs were computed separately from the recovered signal using the proposed approach (Fig. 8(B)) as well as using the fully-sampled data (Fig. 8(A)). We can observe that most of the FOD profiles are well recovered.

The first result is the ability of our method to correctly distinguish multiple fiber populations. The visual comparison clearly shows that using only 34 directions are enough for robustly recovering the fiber orientation function. Thus, the proposed approach preserves the FOD configuration in the sense of fibers number. We found, in fact, that in the single fiber case, the reconstructed FOD correlates to the ground truth very well. It should be noted that the correct fiber orientations were preserved even in the case of the two and three fibers. The algorithm was in most of the cases able to identify the true fiber orientations.

Secondly, we extract the directions of the main fiber population at each voxel. The peaks detection was performed using a global maxima search algorithm on the recovered FOD. The orientation of each peak indicates the orientation of a fiber. This information will be useful in several applications like tractography, segmentation or for visualization of principal diffusion directions. The peaks detection will be also used to compute the second evaluation criteria (angular error). Fig. 9 illustrates that the FOD peaks (Fig. 9(B)) were correctly identified and qualitatively agree with the ground truth (Fig. 9(C)) and the full HARDI acquisition (Fig. 9(A)) with some small angular error due to the noise level (SNR = 30). The angular resolution is, therefore, preserved.

In order to evaluate the resolving power of the proposed approach, experimentation must not be limited to visual aspects. Quantitative experimentation should be, therefore, implemented.

4.2.2. Quantitative evaluation

We quantitatively compare, here, the reconstruction of the FOD function obtained from undersampled data (34 directions) to those with fully-sampled data (65 directions) in the presence of noise.

Rician noise was added to obtain a signal with SNR values equal to 10, 20, 30. The quality of the reconstructions was assessed by using the quality metrics introduced above on the second synthetic phantom involving different number of fiber populations and different inter-fiber angles.

We calculate, then, the success rate (SR) using the three SNR values for the obtained FOD with the full original set directions

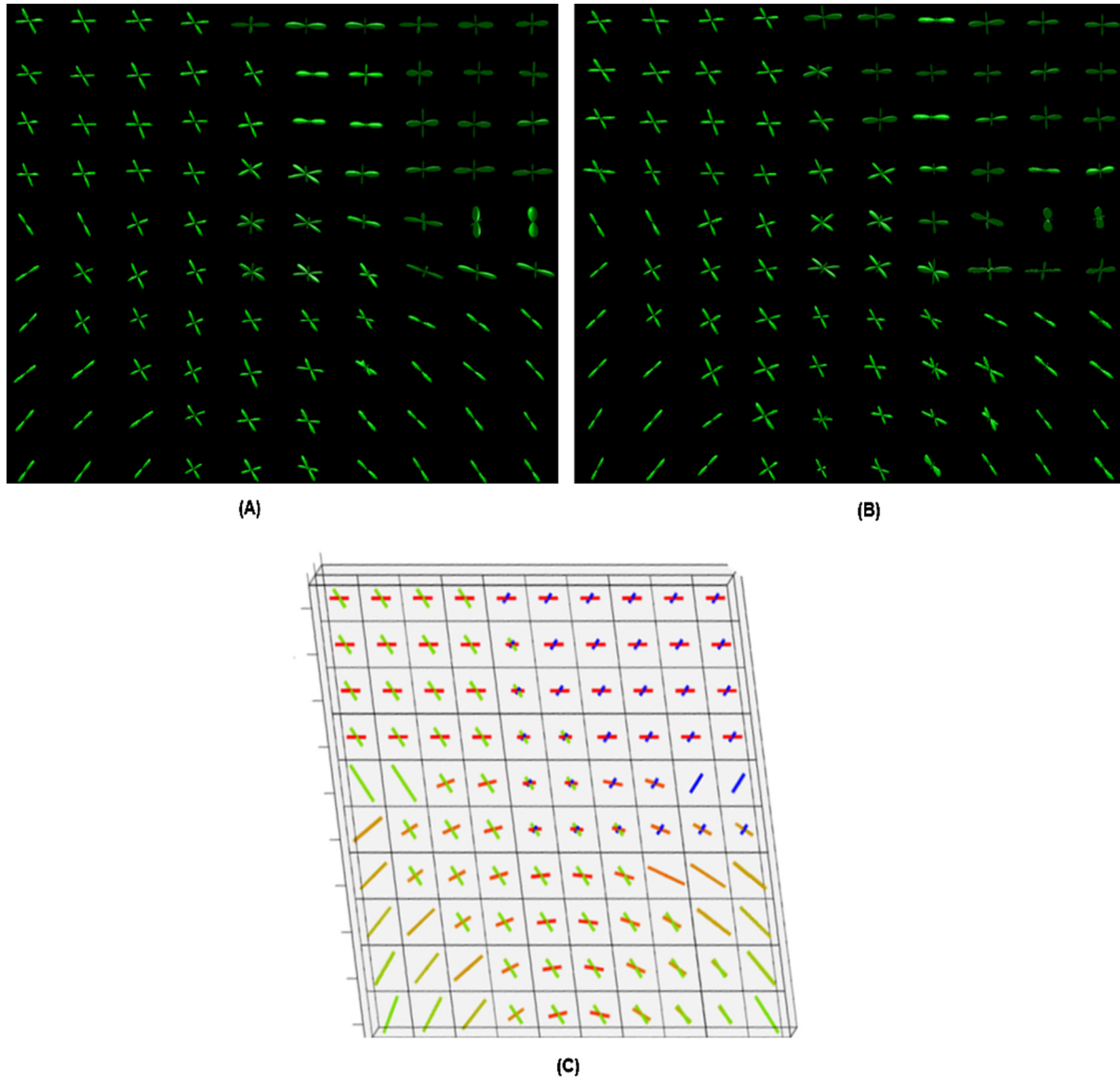


Fig. 8. Qualitative comparison of the obtained results from synthetic noisy data (SNR = 30): (A) the FOD reconstruction using 65 directions, (B) the FOD reconstruction using 34 directions with the proposed approach, (C) the ground truth.

and for the reconstruction of FOD from the estimated HARDI signal using our approach. Fig. 10 shows the good performance and robustness of FOD reconstruction in terms of the number of distinct fiber orientations. In fact, the values of the (SR) criteria are approximately the same for all the values of SNR (10, 20, 30). For example, the success rate using 65 directions was about 61.4% and 61.3% in the case of 34 directions for a fixed SNR = 20. It should be noted that there is a little difference when SNR = 10. This fact can be explained by the effect of the noise.

For the second evaluation criteria, angular error is estimated separately for each voxel, including the original measured data with the full set of 65 gradient directions and its reduced form composed by a subset of 34 directions. We should note that we calculate the angular error by considering only those voxels in phantom where FOD peaks were correctly identified.

We compare the angular error over the entire phantom using all three noise levels. The performance of both methods in term of angular error according to the levels of noise is reported in Fig. 11. It can be seen that the angular error increases as the values of SNR decrease. Moreover, the values of the angular error (θ) based on only 34 directions are approximately the same to those estimated with the full HARDI acquisitions (65 directions) for all the values of SNR (10, 20, 30). For example, the angular error using 65 directions was $3,1^\circ$ and $5,7^\circ$ using 34 directions for a fixed SNR = 20.

As can be appreciated, by using only 34 directions, we can achieve the same reconstruction accuracy when using the full original directions.

The quantitative results confirm the accuracy of the proposed approach as it was demonstrated by the qualitative evaluation. It allows to shorten the DW-MRI acquisition time while keeping accurate fiber orientations estimation.

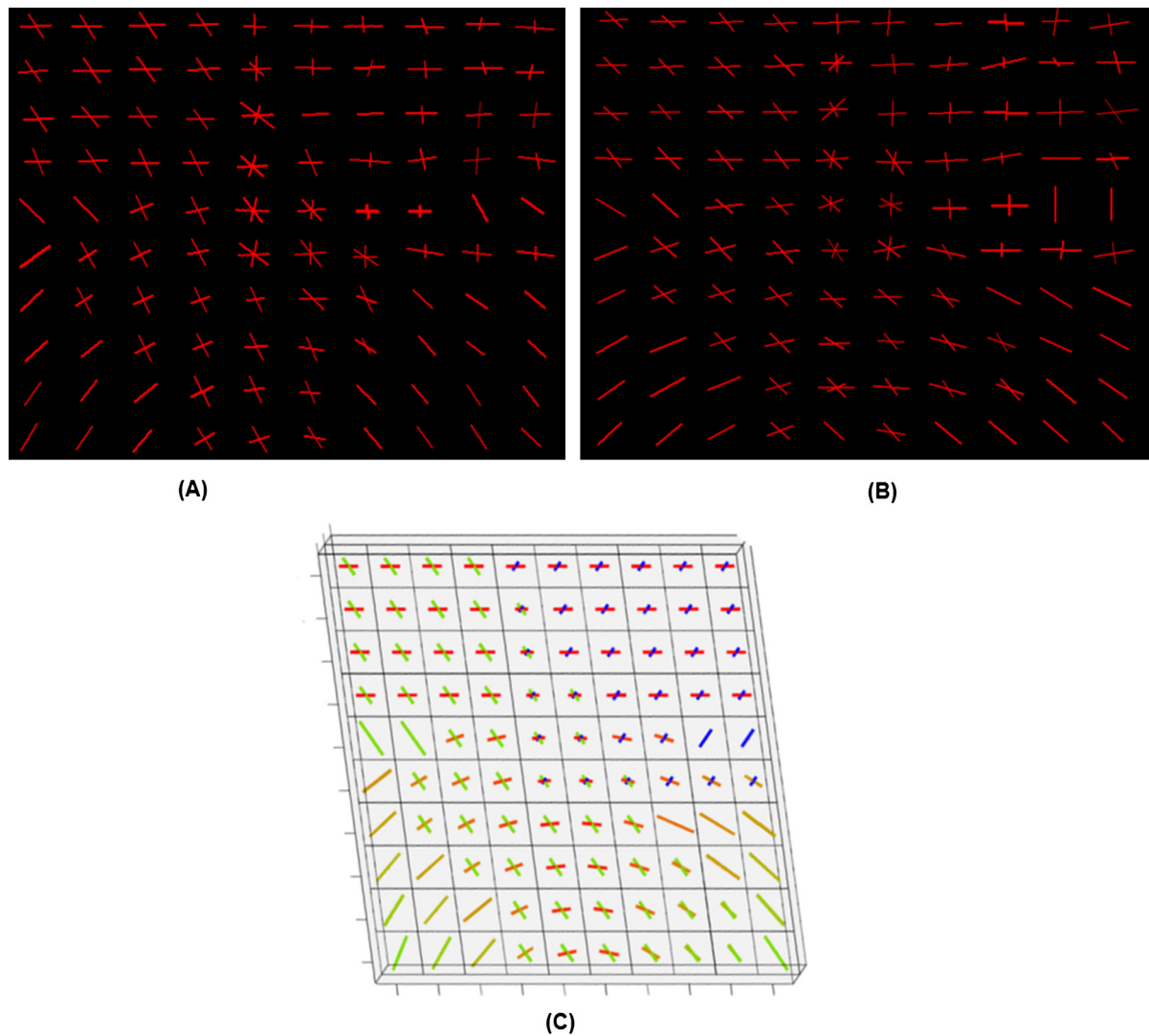


Fig. 9. Main peaks extracted from the FOD estimated in the phantom data for a fixed SNR = 30: (A) 65 directions, (B) 34 directions, (C) the ground truth.

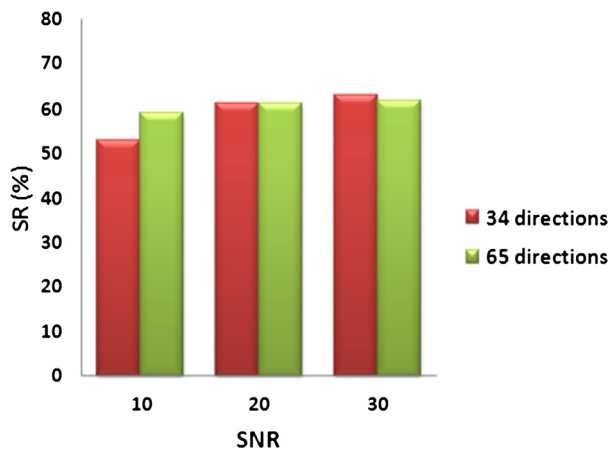


Fig. 10. Comparison of the (SR) criteria values between the original approach and the proposed approach for many noise levels values.

4.3. Comparison framework

For sake of completeness, we also evaluate the effectiveness of our proposed approach in comparison with a large selection

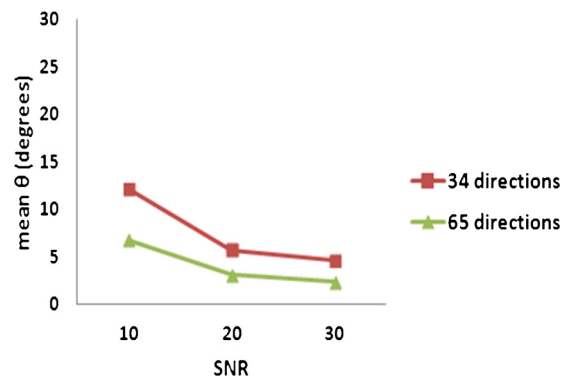


Fig. 11. Comparison of the angular error values between the original approach and the proposed approach according to the noise levels values.

of methods participating in the HARDI reconstruction challenge 2012.³

³ <http://hardi.epfl.ch/static/events/2012ISBI/>.

Table 1
Reconstruction algorithms.

Class	Method	Acquisition parameters
DTI	Diffusion tensor imaging [5]	$b = 1000$, 6 directions
HARDI	QBI, analytical q-ball imaging with the following parameters: $L_{\max} = 6$ and regularization parameter $\lambda = 0.006$ [10]	$b = 3000$, 60 directions
SPARSE	L2-L1-DL, method using dictionary learning in the framework of l2-l1 reconstruction [18]	$b = 2000$, 15 directions
	L2-L1-TV using the l2-l1 problem formulation and including spatial regularization schemes based on total variation [17]	$b = 1200$, 30 directions
	L2-L2 based on l2 norm priors [36]	$b = 3333$, 37 directions
	NN-L2, based on l2 norm priors [37]	$b = 1500$, 48 directions
DSI	DSI-LR, classical DSI enhanced using Lucy–Richardson deconvolution [38]	$b < 8000$, 257 directions

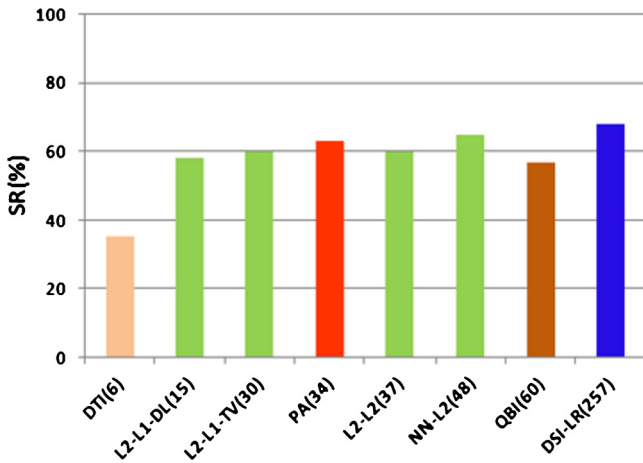


Fig. 12. Comparison of SR between the proposed approach (PA) and different reconstruction methods for a fixed SNR = 30.

Taking the advantage of the public results in [35], we can compare the performance of our proposed approach in terms of success rate and angular error against modern methods from the state-of-the-art. The algorithms have been grouped into four different classes depending on the number of samples used for the reconstruction: DTI-like class, DSI-like class, HARDI-like class and SPARSE-like class. For each class, we choose a single method and for the SPARSE class we used four methods: two CS-based algorithms for recovering HARDI signal from reduced samples (L2-L1-DL and L2-L1-TV) and two methods that are formulated as convex optimization problems and based on the ℓ_2 norm prior. Table 1 shows the selected methods and the acquisition parameters (b -value and number of directions). These methods used the same 3D phantom used in our simulation and the same evaluation criteria described in this paper.

Figs. 12 and 13 show a quantitative comparison of the performance of the proposed approach with the selected methods in terms of SR and angular error.

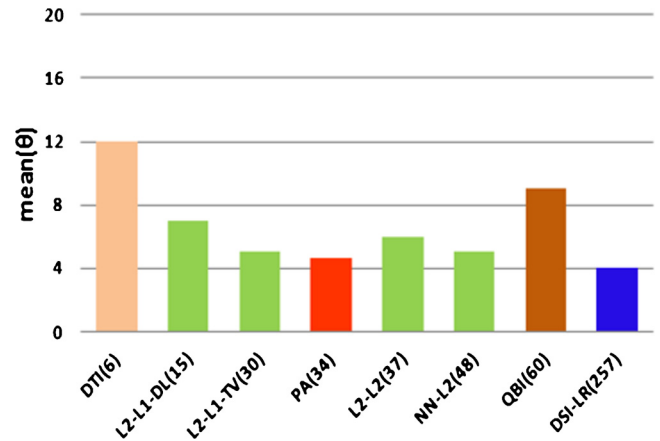


Fig. 13. Comparison of the mean angular error between the proposed approach (PA) and different reconstruction methods for a fixed SNR = 30.

Methods are sorted by the number of samples used for the reconstruction, increasing from left to right. It is shown that the proposed approach can clearly recover the diffusion signal with a good accuracy compared to state of the art approaches even with a small number of measurements. The SR values and the mean angular error were in line with all other SPARSE methods. It should be noted that our method allows to significantly reducing the number of measurements and outperforms the Diffusion Spectrum Imaging (DSI) technique. It requires only 13% the number of measurements needed by DSI technique while maintaining reconstruction quality.

Regarding the four dictionary-based reconstruction approaches inspired by the recent theory of compressed sensing approaches, we can remark that our method gives nearly the same results at an SNR equal to 30 and leads to satisfactory results in terms of accuracy. The proposed approach gives better results in terms of SR and angular error than HARDI method (QBI) that used 60 directions.

5. Conclusion

In this work, we have proposed a novel geometrical approach for recovering the HARDI signal from a reduced number of gradient directions. It is based on a nearest-neighbor linear interpolation approach between the points on the unit sphere of the q-space. The experiments were conducted on two kinds of synthetic noisy diffusion MRI phantoms with known ground-truths. The results are validated by a visual comparison to the corresponding ground truth and to the computed FOD from the full HARDI acquisition (65 directions). We have also quantitatively evaluated and compared the proposed approach in terms of both the correct number of fibers (SR) and the angular error criteria in the presence of noise. Finally, we have compared our approach against methods using the same 3D phantom used in our simulation and the same quality metrics. According to the obtained results of this study, the proposed approach enables accurate FOD reconstruction from a limited number of q-space samples which reduces the scan time significantly while maintaining the data quality. Our novel approach is also robust under the noise.

Future works involve doing extensive validation on several subjects in different brain regions and testing the accuracy of the proposed framework using a hemispherical encoding scheme. It will be also interesting to perform tractography, using fiber orientations estimated by the proposed approach.

Conflict of interest statement

The author has no conflict of interest.

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